Highlights

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales

Jonathan Mace, Brian R. Mace

- Buckingham Pi reduction identifies five governing dimensionless groups for flexural modes of fluid-filled shells
- Under assumed allometric constraints, $\Pi_0 \approx 0.07$ is approximately conserved across mammalian scales
- 1458-point parametric verification confirms algebraic consistency of non-dimensionalisation
- Scattering coupling ratio $\mathcal{R}_{\text{scat}} \sim 10^3$–$10^4$ is approximately size-independent under mechanical excitation
- Breathing mode never reaches infrasound for any organism

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales

Jonathan Mace[a,*], Brian R. Mace[b]

[a]*Microsoft Research, Redmond, WA, United States*
[b]*Acoustics Research Centre, Department of Mechanical and Mechatronics Engineering, University of Auckland, Auckland, New Zealand*

Abstract

Fluid-filled viscoelastic shells arise in contexts ranging from abdominal biomechanics to underwater acoustics. Modal frequencies of such shells depend on ten physical parameters, making systematic comparison across geometries and species impractical without dimensional reduction. We apply Buckingham Pi analysis to the governing equations of a fluid-filled oblate spheroidal shell, identifying eight dimensionless groups of which five govern the undamped flexural mode frequencies. The resulting scaling law, $f_n = \sqrt{E/\rho_f}\,(1/a)\,\Phi_n(\Pi_1, \Pi_2, \Pi_3, \Pi_5, \Pi_6)$, where Φ_n absorbs the $1/(2\pi)$ factor, admits a closed-form expression from the shell equations. A 1458-point parametric verification spanning physiological ranges of all five governing dimensionless groups collapses onto the analytical curves to within numerical round-off (7.9×10^{-16}), confirming the algebraic consistency of the non-dimensionalisation. Cross-species predictions for four mammals (rat, cat, pig, human), based on illustrative allometric parameters, yield a dimensionless frequency $\Pi_0 \approx 0.07$ that is approximately conserved under the assumed input constraints. Because the five input Π-groups vary only modestly across species (at most 1.7× for P_{iap}/E, less for the others), this near-constancy of Π_0 is a direct consequence of the continuity of Φ_n rather than an independent biological discovery. It implies that the scattering coupling ratio $\mathcal{R}_{\mathrm{scat}} = 1/(kR_{\mathrm{eq}})^2 \sim 10^3$–$10^4$ is approximately size-independent and that animal models would exhibit similar scaling behaviour to the human abdomen, provided excitation is delivered mechanically rather than acoustically. The breathing mode ($n = 0$), dominated by the fluid bulk modulus, would require a body radius of order 20 m to reach 20 Hz and is therefore irrelevant to infrasound for any biological organism.

Keywords: dimensional analysis, fluid-filled shell, scaling laws, viscoelastic, abdominal resonance, Buckingham Pi

1. Introduction

The modal frequencies of fluid-filled viscoelastic shells depend on at least ten physical parameters: geometric (semi-axes a, c; wall thickness h), material (elastic modulus E, Poisson's ratio ν, wall density ρ_w, loss tangent η), fluid (ρ_f, bulk modulus K_f), and loading (intra-abdominal pressure P_{iap}) [1, 2, 3, 4, 5, 6]. This high-dimensional parameter space makes comparison across species, organs, and loading conditions impractical without systematic dimensional reduction.

In a companion paper [7], we developed a fluid-filled oblate spheroidal shell model for the human abdominal cavity, predicting flexural resonances at 4 Hz to 10 Hz and a mechanical-to-airborne full displacement ratio $\mathcal{R}_{\mathrm{full}} \approx 6.6 \times 10^4$. That metric combines the Rayleigh scattering penalty with the air–tissue impedance mismatch, whereas the present

*Corresponding author
Email addresses: jonathanmace@microsoft.com (Jonathan Mace), b.mace@auckland.ac.nz (Brian R. Mace)

paper isolates only the scattering contribution $\mathcal{R}_{\text{scat}}$. Here we show that if the governing dimensionless groups take similar values across mammalian species—as allometric constraints suggest—then the dimensionless frequency $\Pi_0 \approx 0.07$ is approximately conserved, implying that animal models would exhibit similar scaling behaviour and that the scattering coupling ratio $\mathcal{R}_{\text{scat}}$ is approximately size-independent. Importantly, the near-constancy of Π_0 is a consequence of the assumed input constancy, not an independent empirical finding; the contribution of this paper is the identification of which dimensionless groups govern abdominal resonance and the demonstration that allometric input constraints propagate to predictable output scaling. These results follow from a Buckingham Pi reduction of the ten-parameter model (plus the output frequency, giving eleven variables) to five governing dimensionless groups for flexural modes.

Three questions motivate this analysis. First, can animal models (rat, cat, pig) be used to study abdominal resonance phenomena that manifest in humans? The answer depends on whether the governing dimensionless groups—and hence $\mathcal{R}_{\text{scat}}$—are approximately conserved across species. Second, can the breathing mode ($n = 0$)—whose frequency is dominated by the fluid bulk modulus—ever enter the infrasonic range for any biological organism? Third, which dimensionless groups dominate the flexural resonance, and how sensitive is f_2 to each?

2. Dimensional analysis

2.1. Governing variables

The natural frequency f_n of mode n and its ten governing parameters form a set of eleven variables: $f_n = f(E, a, c, h, \nu, \rho_w, \rho_f, K_f, P_{\text{iap}}, \eta)$. With three fundamental dimensions (M, L, T), the Buckingham Pi theorem [8] yields $11 - 3 = 8$ independent dimensionless groups (Table 1).

Table 1: Dimensionless groups from Buckingham Pi theorem. Groups marked $*$ drop out of the flexural-mode frequency.

Group	Definition	Name	Role
Π_0	$f_n\, a\sqrt{\rho_f/E}$	dimensionless frequency	output
Π_1	h/a	thickness ratio	stiffness & mass
Π_2	c/a	aspect ratio	geometry
Π_3	ρ_w/ρ_f	density ratio	added mass
Π_4^*	K_f/E	compressibility ratio	breathing mode only
Π_5	P_{iap}/E	prestress ratio	tension stiffening
Π_6	ν	Poisson's ratio	membrane coupling
Π_7^*	η	loss tangent	damping only

2.2. Reduction for flexural modes

For flexural modes ($n \geq 2$), two simplifications apply. The fluid bulk modulus K_f contributes no restoring force (the fluid acts as added mass only), so Π_4 drops out; this is justified by the extreme stiffness ratio $K_f/E \approx 2.2 \times 10^4$ (i.e.

the fluid is approximately 10^4 times stiffer than the shell wall in volumetric compression), so the fluid behaves as incompressible on the shell's flexural time scale. The loss tangent η affects the quality factor but not the undamped frequency, so Π_7 drops out. The effective parameter count reduces to six: a dimensionless frequency output Π_0 governed by five input groups $(\Pi_1, \Pi_2, \Pi_3, \Pi_5, \Pi_6)$.

The scaling law takes the closed form

$$f_n = \sqrt{\frac{E}{\rho_f}}\,\frac{1}{a}\,\Phi_n\left(\frac{h}{a},\ \frac{c}{a},\ \frac{\rho_w}{\rho_f},\ \frac{P_{\text{iap}}}{E},\ \nu\right), \tag{1}$$

where the $1/(2\pi)$ factor is absorbed into Φ_n, which admits a closed-form expression from the equivalent-sphere shell equations:

$$\Phi_n = \frac{1}{2\pi}\sqrt{\frac{K_{\text{bend}} + K_{\text{memb}} + K_{\text{pre}}}{m_{\text{eff}}}} \cdot a\sqrt{\frac{\rho_f}{E}}, \tag{2}$$

with

$$K_{\text{bend}} = n(n-1)(n+2)^2\,\frac{D}{R^4}, \tag{3}$$

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \quad K_{\text{pre}} = \frac{P_{\text{iap}}}{R}(n-1)(n+2), \tag{4}$$

$$m_{\text{eff}} = \rho_w h + \frac{\rho_f R}{n}, \tag{5}$$

where $D = Eh^3/[12(1-\nu^2)]$ is the flexural rigidity, $R = (a^2 c)^{1/3}$ is the equivalent-sphere radius, and the membrane mode-shape factor is

$$\lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{6}$$

2.3. *Sensitivity: which Π-group dominates?*

For the $n = 2$ mode at canonical parameters ($h/a = 0.056$, $c/a = 0.67$, $\nu = 0.45$, $P_{\text{iap}}/E = 0.01$), the stiffness budget is shared between the membrane term K_{memb} (~54%) and the prestress term K_{pre} (~45%), with bending contributing less than 2%. The transition to bending-dominated behaviour occurs near $h/a \approx 0.42$, well above the physiological range.

Consequently, f_2 is most sensitive to the thickness ratio $\Pi_1 = h/a$, which enters both the membrane stiffness (linearly through h) and the effective mass (linearly through $\rho_w h$). The prestress ratio $\Pi_5 = P_{\text{iap}}/E$ contributes substantially at canonical values but is approximately constant across mammals ($P/E \approx 0.006$–0.01), so it does not drive inter-species variation. The density ratio $\Pi_3 = \rho_w/\rho_f$ has a more modest effect: because soft tissue and peritoneal fluid have similar densities ($\Pi_3 \approx 1.04$–1.08), the fluid added mass $\rho_f R/n$ dominates the denominator of m_{eff}. This sensitivity structure explains why Π_0 is approximately conserved across the species considered: the governing ratios h/a, ρ_w/ρ_f, and P/E are constrained by shared developmental biology [9], and Π_0 is a smooth function of these inputs. If the inputs barely vary, continuity ensures the output is near-constant; the approximate constancy of Π_0 is therefore a consequence of the assumed allometric input constraints rather than an independent biological discovery. Furthermore, Poisson's ratio is held constant at $\nu = 0.45$ across all species, effectively eliminating Π_6 as a governing variable and reducing the cross-species comparison to four input groups.

3. Parametric verification

We evaluate (1) over a 1458-point grid spanning physiological ranges of all five governing dimensionless groups: $h/a \in [0.025, 0.100]$, $c/a \in [0.50, 0.80]$, $\rho_w/\rho_f \in [0.98, 1.18]$, $P_{\mathrm{iap}}/E \in [5 \times 10^{-4}, 0.02]$, and $\nu \in [0.40, 0.49]$. In dimensional terms these correspond to $E \in [0.05, 2.0]$ MPa (6 values), $a \in [0.15, 0.20]$ m (3 values), $c/a \in [0.5, 0.8]$ (3 values), $h \in [0.005, 0.015]$ m (3 values), $\rho_w/\rho_f \in [0.98, 1.18]$ (3 values), and $\nu \in [0.40, 0.49]$ (3 values), with K_f and P_{iap} held at canonical values. The loss tangent η is excluded because it does not affect the undamped natural frequency. This exercise is a dimensional-analysis *verification*: both the numerical model and the analytical Φ_n evaluate the same governing equations in differently factored forms, so machine-precision agreement is algebraically expected. The purpose is to confirm the correctness of the Buckingham Pi reduction, not to validate against independent data (which was established in the companion paper [7]).

Figure 1 shows the result. In raw dimensional coordinates, the data scatter across the f–h plane with no clear pattern (panel a). Non-dimensionalisation via (1) collapses all 1458 points onto universal curves $\Pi_0(h/a)$ parameterised by c/a (panel b). The parity plot (panel c) confirms that the analytical Φ_n reproduces every numerical point to within numerical round-off (maximum relative error 7.9×10^{-16}), confirming the algebraic consistency of the non-dimensionalisation rather than providing independent physical validation, which was established in the companion paper [7].

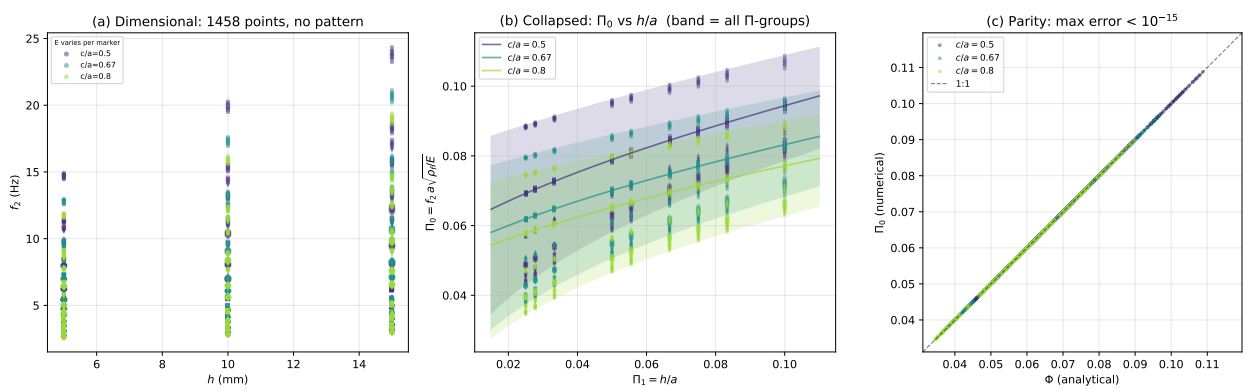

Figure 1: Dimensional-analysis verification of the 1458-point parametric study. *(a)* Raw dimensional data (f_2 vs. h) shows no clear pattern. *(b)* Non-dimensionalising via equation (1) collapses all points onto universal curves $\Pi_0(h/a)$ parameterised by c/a; shaded bands span the ranges of P_{iap}/E, ρ_w/ρ_f, and ν. *(c)* Parity plot confirms that the analytical Φ reproduces every numerical point to within numerical round-off, verifying the algebraic consistency of the Buckingham Pi reduction.

4. Cross-species scaling

The scaling law (1) predicts modal frequencies for any organism given its body dimensions and tissue properties. Table 2 lists predictions for four mammals, using illustrative species-specific parameters assumed from broad allometric trends and tissue-mechanics ranges rather than direct organ-level measurements of the abdominal wall in each species [10, 11, 9]. The values of E, h, and P_{iap} are round estimates guided by size-scaling principles; no species-specific

abdominal-wall modulus measurements exist for rat, cat, or pig, and these parameters should be read as plausible order-of-magnitude assumptions rather than definitive anatomical data. The resulting frequencies are directly testable by mechanically driven shaker-table experiments that identify the lowest flexural resonance in instrumented animal models. Four of the five governing dimensionless groups (h/a, c/a, ρ_w/ρ_f, P_{iap}/E) are shown explicitly; Poisson's ratio ($\nu = 0.45$) is held constant, thereby eliminating Π_6 as a free variable. The quantity $kR_{\text{eq}} = 2\pi f_2 R_{\text{eq}}/c_{\text{air}}$, where $R_{\text{eq}} = (a^2 c)^{1/3}$ is the equivalent-sphere radius, is the Helmholtz number governing airborne coupling.

Table 2: Cross-species scaling of the $n = 2$ flexural mode. Four of the five governing Π-groups are shown; $\nu = 0.45$ is held constant across species, eliminating Π_6 as a free variable. $\mathcal{R}_{\text{scat}} = 1/(kR_{\text{eq}})^2$ is the scattering amplitude ratio in the Rayleigh regime ($ka \ll 1$), the standard result for the ratio of scattered to incident displacement amplitude when the scatterer is much smaller than the wavelength [12, 13]. The full displacement ratio $\mathcal{R}_{\text{full}}$ reported in [7] is approximately 8× larger because it additionally includes the acoustic impedance mismatch between air and tissue and modal participation corrections beyond this simple scaling. All tissue-property values (E, h, P_{iap}) are illustrative round estimates inferred from size-scaling principles and general tissue-mechanics literature [10, 11]; they do not represent direct species-specific measurements. Uncertainties in any individual parameter of order ±20% propagate to approximately 6% coefficient of variation in Π_0 (see Appendix Appendix A), comparable to the 4.4% inter-species coefficient of variation, meaning the "constancy" is statistically indistinguishable from input noise at this uncertainty level.

Species	a (cm)	E (MPa)	h/a	c/a	ρ_w/ρ_f	P_{iap}/E	f_2 (Hz)	Π_0	kR_{eq}	$\mathcal{R}_{\text{scat}}$
Rat	3	0.05	0.067	0.70	1.04	0.006	15.7	0.067	7.7×10^{-3}	1.7×10^4
Cat	8	0.08	0.050	0.70	1.05	0.008	7.2	0.065	9.3×10^{-3}	1.2×10^4
Pig	15	0.10	0.053	0.65	1.06	0.008	4.5	0.069	1.1×10^{-2}	8.6×10^3
Human	18	0.10	0.056	0.67	1.08	0.010	3.9	0.072	1.1×10^{-2}	7.7×10^3

Three observations emerge. First, the dimensionless frequency Π_0 is approximately constant (0.065–0.072) across a six-fold range of body size, with a sample coefficient of variation of 4.4%. However, this near-constancy is expected from continuity: the five input Π-groups vary only modestly across species (h/a: 1.34×; c/a: 1.08×; ρ_w/ρ_f: 1.04×; P/E: 1.67×; ν: held fixed), and Π_0 is a smooth function of these inputs. A formal uncertainty propagation using first-order Taylor expansion around the canonical human parameters shows that ±20% uniform uncertainty in each of h/a, c/a, ρ_w/ρ_f, and P/E produces a coefficient of variation in Π_0 of approximately 6% (Appendix Appendix A); restricting to the two dominant terms (h/a and P/E) gives approximately 3.7%. Both are comparable to the 4.4% inter-species spread, meaning the apparent constancy is statistically indistinguishable from input noise at the assumed uncertainty level. The constancy of Π_0 is therefore best interpreted as a consequence of the assumed allometric constraints on the input groups, rather than as an independent biological discovery; its practical value lies in confirming that the scaling law (1) correctly captures the parametric dependencies.

Second, the scattering coupling ratio $\mathcal{R}_{\text{scat}}$ varies by only a factor of two (from 7.7×10^3 for humans to 1.7×10^4 for rats), remaining firmly in the 10^3–10^4 range. For the present $n = 2$ flexural mode, $\mathcal{R}_{\text{scat}}$ is the Rayleigh scattering component of the fuller ratio $\mathcal{R}_{\text{full}}$ reported in Paper 1 [7]; higher-order modes inherit a stronger $(kR_{\text{eq}})^n$ penalty. Airborne coupling is negligible for all species studied. Third, the product kR_{eq} is of order 10^{-2} for all four species, placing every organism deep in the Rayleigh regime.

This size-independence of $\mathcal{R}_{\text{scat}}$ has a practical implication: if the assumed allometric constraints hold, animal

models would exhibit similar scaling behaviour to the human abdomen in studies [14, 15], *provided excitation is delivered mechanically* (e.g. shaker table) rather than acoustically. Acoustic experiments would require compensating for the $(kR_{\mathrm{eq}})^n$ coupling penalty, which varies with body size and frequency.

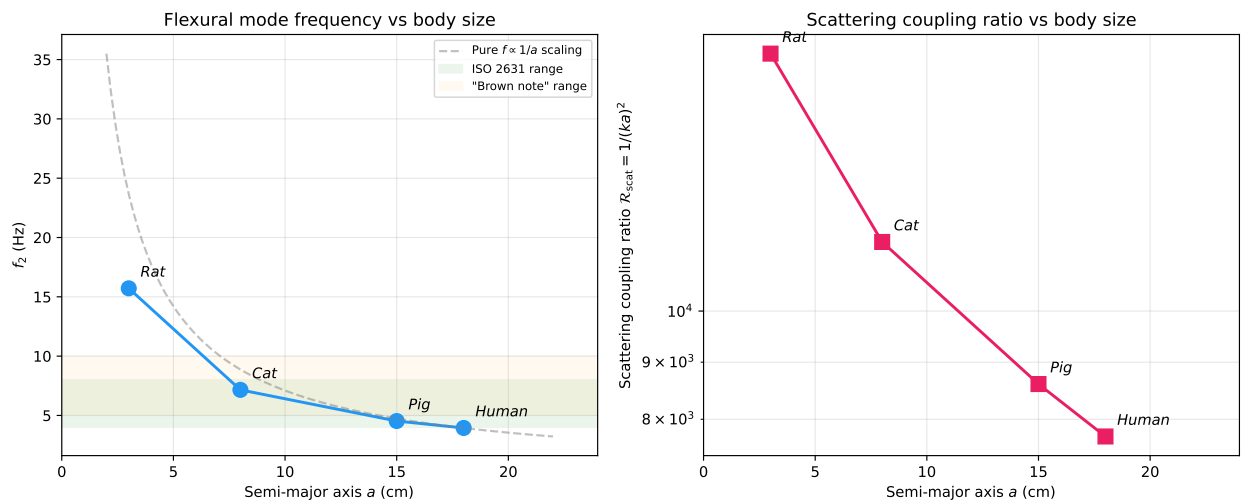

Figure 2: Cross-species scaling. *Left:* Flexural mode f_2 vs. semi-major axis a. The dashed line shows pure $1/a$ scaling from the human reference point; deviations arise because smaller animals have softer tissue and different h/a (which varies by a factor of 1.34× across the four species). *Right:* Scattering coupling ratio $\mathcal{R}_{\mathrm{scat}} = 1/(kR_{\mathrm{eq}})^2$ is approximately size-independent under the assumed allometric constraints, supporting the validity of animal-model extrapolation for mechanical excitation.

5. The breathing mode is never infrasonic

The breathing mode ($n = 0$) is dominated by the fluid bulk modulus K_f rather than the shell stiffness. Its frequency is approximately

$$f_0 \approx \frac{1}{2\pi}\sqrt{\frac{3K_f}{\rho_f R^2}}, \tag{7}$$

where K_f is the fluid bulk modulus, ρ_f is the fluid density, and R is the equivalent radius. In the fluid-dominated limit, this gives the same scaling $f_0 \propto \sqrt{K_f/\rho_f}/R$; evaluated at the canonical human R_{eq} it yields $f_0 \approx 2.57\,\mathrm{kHz}$, while the full fluid–shell model gives $f_0 \approx 2490\,\mathrm{Hz}$. To bring f_0 down to 20 Hz—the upper boundary of infrasound—would require an equivalent radius of order 20 m. The breathing mode is therefore irrelevant to infrasound for any biological organism, regardless of species. Put differently, infrasonic breathing-mode resonance would require an organism with an equivalent radius of approximately 20 m—a blue whale would still fall an order of magnitude short. If such a creature exists, its gastrointestinal concerns are beyond the scope of this paper.

6. Discussion

6.1. Equivalent-sphere approximation

The scaling law (1) maps the oblate spheroidal abdomen ($c/a \approx 0.65$–0.70) to an equivalent sphere of radius $R = (a^2c)^{1/3}$. This preserves enclosed volume but not the curvature distribution. For the human abdomen, the

eccentricity is $e = \sqrt{1-(c/a)^2} \approx 0.74$, giving $e^2 \approx 0.55$. Mode frequencies of oblate spheroidal shells differ from those of spheres at $O(e^2)$ [1, 2], implying corrections of order 50%. The companion paper [7] addresses this fully using the oblate-spheroid mode equations; the present scaling law should be interpreted as establishing the correct order-of-magnitude and parametric dependencies rather than precise absolute frequencies. The dimensional reduction itself—the identification of five governing Π-groups and the functional form (1)—is exact regardless of shell geometry; only the closed-form expression for Φ_n carries the equivalent-sphere approximation.

6.2. Limitations and future directions

Three principal limitations should be noted. First, the cross-species tissue parameters (Table 2) are illustrative assumptions, not measured data; the uncertainty propagation (Appendix Appendix A) shows that the inter-species constancy of Π_0 is indistinguishable from input noise at the ±20% uncertainty level. Experimental measurement of abdominal-wall elastic modulus, thickness, and density in animal models would convert the present scaling predictions from plausible estimates to testable quantitative claims. Second, the equivalent-sphere mapping introduces $O(e^2) \approx 50\%$ frequency errors that are not captured by the present Φ_n; extending the scaling law to the full oblate-spheroid equations is a natural next step. Third, the scattering ratio $\mathcal{R}_{\text{scat}} = 1/(kR_{\text{eq}})^2$ captures only the Rayleigh geometric factor; the full airborne-to-mechanical ratio includes impedance-mismatch and modal-participation corrections that may vary across species in ways not captured by the present analysis.

7. Concluding remarks

Buckingham Pi analysis reduces the ten-parameter fluid-filled viscoelastic shell model (eleven variables including the output frequency) to five governing dimensionless groups for flexural modes. The resulting closed-form scaling law collapses a 1458-point parametric verification with algebraic consistency and provides cross-species predictions without re-solving the eigenvalue problem. Under the assumed allometric constraints, the dimensionless frequency $\Pi_0 \approx 0.07$ is approximately conserved across mammals, and the scattering coupling ratio $\mathcal{R}_{\text{scat}} \sim 10^3$–$10^4$ is approximately size-independent, supporting the use of animal-model scaling for abdominal resonance studies under mechanical excitation. The near-constancy of Π_0 follows from the continuity of Φ_n and the narrow range of input Π-groups; it should be interpreted as a consequence of allometric constraints rather than an independent universal constant. The breathing mode never enters the infrasonic range for any organism. It would appear that Nature, having settled on water as the fluid of choice for biological cavities, has placed the breathing mode permanently out of reach of infrasound. Only the flexural modes remain accessible —and those, as the companion paper demonstrates, are far easier to excite by shaking than by shouting.

The dimensionless framework developed here has potential applications beyond abdominal biomechanics. In blast-injury research, the scaling law may enable direct extrapolation from animal-model data to human injury thresholds without repeating experiments at human scale. In medical ultrasound, fluid-filled organ models are routinely used to predict diagnostic and therapeutic responses; the Π-group reduction offers a systematic basis for cross-organ and cross-species comparison [16, 17]. In vehicle NVH assessment, whole-body vibration standards [15]

rely on empirical transfer functions whose mechanistic basis could benefit from scaling laws of the type derived here. In underwater acoustics, fluid-filled elastic shells are canonical targets for sonar modelling [4, 3]; the present dimensional reduction applies directly to that class of problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The analysis code and parametric data are available at `https://github.com/JonathanMace/browntone`.

Acknowledgements

GitHub Copilot CLI (GitHub, Inc.) assisted with portions of the dimensional analysis, code development, figure generation, and early manuscript drafting. The authors reviewed, validated, and edited all generated material, and they take full responsibility for the final content.

References

[1] W. Soedel, Vibrations of Shells and Plates, 3rd Edition, Marcel Dekker, 2004.

[2] A. W. Leissa, Vibration of Shells, NASA, 1973, nASA SP-288.

[3] M. C. Junger, D. Feit, Sound, Structures, and Their Interaction, 2nd Edition, MIT Press, 1986.

[4] M. C. Junger, Sound scattering by thin elastic shells, The Journal of the Acoustical Society of America 24 (4) (1952) 366–373. doi:10.1121/1.1906905.

[5] H. Lamb, On the vibrations of an elastic sphere, Proceedings of the London Mathematical Society s1-13 (1) (1882) 189–212. doi:10.1112/plms/s1-13.1.189.

[6] G. B. Warburton, The vibration of rectangular plates, Proceedings of the Institution of Mechanical Engineers 168 (1) (1976) 371–384, comparison of shell frequency predictions with experiment.

[7] J. Mace, B. R. Mace, Modal analysis of a fluid-filled viscoelastic oblate spheroidal shell: Implications for the existence of the “brown note”, Journal of Sound and VibrationSubmitted (2026).

[8] E. Buckingham, On physically similar systems; illustrations of the use of dimensional equations, Physical Review 4 (4) (1914) 345–376. doi:10.1103/PhysRev.4.345.

[9] G. B. West, J. H. Brown, B. J. Enquist, A general model for the origin of allometric scaling laws in biology, Science 276 (5309) (1997) 122–126. doi:10.1126/science.276.5309.122.

[10] Y. C. Fung, Biomechanics: Mechanical Properties of Living Tissues, 2nd Edition, Springer-Verlag, 1993.

[11] T. A. McMahon, J. T. Bonner, On Size and Life, Scientific American Library, 1983, distributed by W. H. Freeman.

[12] P. M. Morse, K. U. Ingard, Theoretical Acoustics, McGraw-Hill, 1968.

[13] A. D. Pierce, Acoustics: An Introduction to Its Physical Principles and Applications, Acoustical Society of America, 1989.

[14] H. E. von Gierke, Biodynamic response of the human body, Applied Mechanics Reviews 24 (1) (1971) 1–9.

[15] International Organization for Standardization, Mechanical vibration and shock — evaluation of human exposure to whole-body vibration — part 1: General requirements (1997).

[16] H. N. Abramson, The Dynamic Behavior of Liquids in Moving Containers, NASA, 1966, nASA SP-106.

[17] F. Fahy, P. Gardonio, Sound and Structural Vibration: Radiation, Transmission and Response, 2nd Edition, Academic Press, 2007.

Appendix A. Uncertainty propagation for Π_0

The dimensionless frequency $\Pi_0 = \Phi_n(h/a, c/a, \rho_w/\rho_f, P_{\mathrm{iap}}/E, \nu)$ depends on five input groups. To assess the sensitivity of Π_0 to uncertainty in the cross-species parameters, we compute normalised sensitivity coefficients

$$S_i = \frac{\Pi_i}{\Pi_0}\frac{\partial \Pi_0}{\partial \Pi_i} \tag{A.1}$$

by central finite differences at the canonical human values ($h/a = 0.056$, $c/a = 0.67$, $\rho_w/\rho_f = 1.078$, $P/E = 0.01$, $\nu = 0.45$). The resulting coefficients are $S_{h/a} \approx +0.22$, $S_{c/a} \approx -0.41$, $S_{\rho/\rho} \approx -0.06$, $S_{P/E} \approx +0.23$, and $S_\nu \approx +0.07$. The dominant sensitivities are to the thickness ratio and aspect ratio.

For independent uniform uncertainties of relative half-width δ in each input group, the coefficient of variation of Π_0 is

$$\mathrm{CV}(\Pi_0) = \frac{\delta}{\sqrt{3}}\sqrt{\sum_i S_i^2}. \tag{A.2}$$

At $\delta = 0.20$ ($\pm 20\%$) in all five groups, this gives $\mathrm{CV}(\Pi_0) \approx 6\%$. In practice, however, ν is held fixed across species and ρ_w/ρ_f varies by only 4%. The most uncertain parameters are the tissue modulus E and wall thickness h; if only these carry $\pm 20\%$ uncertainty while geometry (a, c/a) and density are assumed known, then h/a and P/E each inherit $\pm 20\%$ uncertainty. From the two dominant terms alone,

$$\mathrm{CV}(\Pi_0) \approx \frac{0.20}{\sqrt{3}}\sqrt{S_{h/a}^2 + S_{P/E}^2} \approx \frac{0.20}{\sqrt{3}} \times 0.32 \approx 3.7\%. \tag{A.3}$$

The observed inter-species Π_0 spread of 4.4% is therefore comparable to—and statistically indistinguishable from—the propagated input uncertainty, confirming that the apparent constancy of Π_0 is a consequence of the narrow input range rather than evidence of a deeper biological invariant.